BETSEBAY GETAHUN

Addis Ababa, Ethiopia • getahunbetsebay63@gmail.com • +251 90 825 6262 • github.com/betsebaygetahun

EDUCATION

Addis Ababa University — Addis Ababa, Ethiopia
Bachelor of Science in Computer Engineering — Expected June 2027

- CGPA: Above 3.5 / 4.0
- Relevant Coursework: Database Systems, Data Structures & Algorithms, Computer and Network Security, Data Communication and Computer Networks, Software Engineering, Computer Architecture and Organization, Microcomputer and Interfacing, Digital Signal Processing

HONORS & AWARDS

6th International Olympiad on Financial Security — Krasnoyarsk, Russia
National Finalist – Representing Ethiopia — September 2026 – October 2026

- Selected to represent Ethiopia on the international stage at Siberian Federal University, competing against ~700 school and university students from 40+ countries in advanced financial security, cryptography, and risk analysis challenges.

TECHNICAL PROJECTS

Restaurant Operations Management System (ROMS) — Addis Ababa, Ethiopia
Lead / Independent Flutter Developer — June 2026 – Present

- Built ROMS from scratch — a full restaurant operations platform in Flutter 3 & Firebase with 5 role-based interfaces (Admin, Manager, Waiter, KDS, Cashier) handling real orders in a live commercial environment.
- Chose Clean Architecture with Riverpod 2 and fpdart to keep 16 business use-cases modular and crash-free under heavy async load — a deliberate design decision that made adding new features later far less risky.
- Solved a billing race condition by wrapping order settlement, table reallocation, and tax computation inside atomic Firestore transactions, locking in 99.9% data consistency across concurrent operations.

Ethiopian Youth Council Platform — Addis Ababa, Ethiopia
Full-Stack / Lead Frontend Developer – Team Project — April 2026 – June 2026

- Led the frontend of a national youth civic platform — bilingual (Amharic/English) from day one using next-intl, built on Next.js 16 (App Router), React 19, and TypeScript, serving regional councils across Ethiopia.
- Designed and coded an interactive SVG map of Ethiopia using ethiopia-geo-boundaries and Framer Motion, letting users click into any region to explore its leadership, membership, and local programs.
- Built the CMS that powers the whole platform — role-gated (Super Admin, Regional Admin, Editor), with a Tiptap rich-text editor and Zod-validated forms for managing news, events, and audit-logged activity.

Tef-Tef Freelance Marketplace Platform — Addis Ababa, Ethiopia
Full-Stack Developer – Team Project — March 2026 – June 2026

- Contributed full-stack to Tef-Tef, a cross-platform freelance marketplace — building the web frontend (React), mobile app (Flutter), and parts of the Node.js/TypeScript backend with JWT and Google OAuth2 authentication.
- Integrated Chapa, Ethiopia's leading payment gateway, to power an in-app wallet where users deposit ETB, buy "Connects" credits, and track transactions — the feature that made the platform commercially viable.
- Built the real-time layer with Socket.IO, connecting Flutter and React clients so employers and freelancers get instant chat, live presence indicators, and proposal notifications without refreshing the page.

Dinkenesh Event Management System (DEMS) — Addis Ababa, Ethiopia
Full-Stack Developer – Team Project — July 2026 – August 2026

- Co-built DEMS with a team — a full event lifecycle platform using React 19, Vite, Express 5, and Prisma on PostgreSQL, covering everything from event creation and ticketing to organizer payouts.
- Built the QR-based gate scanner using jsQR — fast enough to clear real venue queues, with scan-event logging that catches duplicate ticket fraud before anyone walks through the door.
- Designed the permission system for 4 user roles (Attendees, Verified Organizers, Gate Security, Platform Moderators) with audit logs that made accountability traceable across every action on the platform.

VoiceGuard – Real-Time AI Voice Scam & Deepfake Detector — Addis Ababa, Ethiopia
AI & Full-Stack Developer — August 2026 – September 2026

- Built VoiceGuard to address a problem I found genuinely alarming — AI voice cloning being used to scam people on live calls. The platform detects deepfake voices and social engineering patterns within a sub-6-second latency window.

- Designed a dual-stream pipeline running Deepgram nova-3 STT and Resemble AI DETECT-3B simultaneously, so the system transcribes speech and scores voice authenticity in parallel, then fuses both signals through a rule-based lexicon engine.
- Added Google Gemini Multimodal as a cross-check layer, feeding it raw audio to catch manipulation patterns the statistical models miss — with graceful fallbacks so the system keeps running even if Gemini is unavailable.

TECHNICAL SKILLS

Languages:
- JavaScript, TypeScript, Python, Java, Dart, Kotlin, HTML5/CSS3

Frontend Frameworks:
- React 19, Next.js 16 (App Router, RSC)
- Vue.js, Vite, Tailwind CSS v4
- shadcn/ui, Radix UI, Framer Motion
- Lucide Icons, next-themes, next-intl

Mobile Development:
- Flutter 3 (Web & Mobile)
- React Native
- Riverpod 2, BLoC, GoRouter 14
- Flutter ScreenUtil, Dio, GetIt

Architecture & Patterns:
- Clean Architecture, MVVM
- Repository Pattern, Dependency Injection
- Functional Programming (fpdart)
- Microservices, RESTful API Design

Tools & Practices:
- Git / GitHub
- GitHub Actions (CI/CD)
- Agile / Scrum, Jira
- Figma, Postman, Docker
- Linux / Bash, VS Code, IntelliJ IDEA
- Linting & Static Analysis

Backend & APIs:
- Node.js, Express.js / Express 5
- Socket.IO (WebSockets), GraphQL
- Streaming Architectures, Multer
- PDFKit, Zod, React Hook Form
- Tiptap WYSIWYG Editor

Databases & Storage:
- Firebase Firestore (Atomic Transactions)
- PostgreSQL, MySQL, MongoDB, Redis
- Prisma ORM 7, Mongoose
- SharedPreferences, Flutter Secure Storage

Cloud & DevOps:
- Firebase (Auth, FCM, Storage, Crashlytics)
- Vercel, Cloudinary API
- NextAuth / JWT, Google OAuth2
- Apple Sign-In, Chapa API (ETB)
- Dynamic SEO Sitemaps & Metadata

AI & Audio Processing:
- Deepgram API (nova-3 Streaming STT)
- Resemble AI (DETECT-3B)
- Google Gemini Multimodal Audio API
- Web Audio API, Score Fusion Algorithms

Special Interests:
- Distributed systems & system design
- Cybersecurity & deepfake detection
- Cloud computing & enterprise apps
- UI/UX design, DevOps & CI/CD

LANGUAGES

Native: Amharic

Professional: English

INTERESTS

Human psychology literature, digital content strategy, video editing, UI/UX graphic design